

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SP8 – General Insurance: Pricing

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** a) State any four principles of insurance with examples. [4]
- b) Explain with reasons if reimbursement of maternity expenses a valid form of insurance. [2]
- [6 Marks]**

Q. 2) Company XYZ is an insurance company commencing to underwrite commercial property risks. It approaches reinsurer ABC to mitigate its risk via Surplus form of Reinsurance with INR 500,000 retention limit.

- a) Calculate cession %, Loss Ratio for insurance company & viability of reinsurance cover for each policy mentioned below:

Policy number	Sum Insured	Premium amount	Claim amount
Policy 1	750,000	15,000	13,000
Policy 2	1,000,000	20,000	25,000

[7]

- b) Reinsurer ABC is trying to use retrocede for this arrangement. Define what is meant by Retrocession. [1]

[8 Marks]

Q. 3) An individual has recently opened up an animal rescue centre called “Forest Star”. He is worried about the size of his investment and associated risks attached, post opening it up for tourists.

- a) List the perils that “Forest Star” could be exposed to. [2]
- b) List the commonly available insurance products to insure the perils identified above. [2]
- c) List the key risk factors for insuring the identified perils. [3]
- d) Explain the difference between risk factors & rating factors. [1]
- e) Suggest with reasons any four of the above listed risk factors that can be used as rating factors. [4]

[12 Marks]

- Q. 4)** a) Define a captive insurance Company giving an example in the context of Indian insurance market. [3]
- b) Suggest main reasons for establishing a captive insurance company. [3]
- [6 Marks]**

Q. 5) A Pharma major corporate house is seeking an opportunity to enter the general insurance market in India. Being new to this industry, they have sought suggestions, with examples, on the regulatory restrictions encountered on the actions of a non-life insurer in India.

[10 Marks]

Q. 6) You are the pricing actuary of a General Insurance company. You had recently launched special insurance towards low-speed E-bikes used by Gig Economy workers. On review of performance after 1 year, the product has made heavy losses. Preliminary analysis shows “Economic factors” have led to the sub optimal performance.

Describe in detail, which factors could have resulted in such performance of this product?

[5 Marks]

- Q. 7)** The market for property insurance in India has undergone a significant change. All insurers are now underwriting policies on technical prices only. You had used traditional burn cost approach on your data to derive your technical prices. On analysing your portfolio after a few months post this change, the Company is still making losses.

Explain to your CEO despite why Burn Cost approach pricing didn't work?

[5 Marks]

- Q. 8)** You have suggested to recruit an actuarial analyst within the reinsurance department of the Company. The Head HR asks you to justify the value addition of this resources in the Reinsurance department.

- a)** State the tasks expected to be performed by the actuarial analyst.

[4]

The Head HR hesitantly approves the hiring with a condition to re-evaluate after 12-months. Below are the performance details of the Marine treaty structure (in the order of cession) as suggested by the actuarial student.

Quota share cession - 15%

Risk XoL – INR 50,00,000 x/s 20,00,000

Aggregate XoL – INR 50,00,000 x/s 25,00,000

Claim No.	Gross Loss (INR)
Claim 1	50,00,000
Claim 2	10,00,000
Claim 3	3,50,000
Claim 4	1,50,000

- b)** Determine the recovery made under each of the above treaties.

[6]

Company would like to reduce the Risk XoL treaty to INR 10,00,000 x/s INR 5,00,000 and purchase Facultative Risk XoL only for Claim 1 at INR 50,00,000 x/s INR 20,00,000.

- c)** Suggest with reasons if the above reinsurance arrangement would benefit the Company.

[3]
[13 Marks]

- Q. 9)** You are pricing a Pet Insurance product for your Company using standard GLM method. The following information is available to you:

Base Frequency	10%
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Base Severity	50,000
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Relativity tables

Animal	Frequency	Severity
Dog	1.00	0.95
Cat	0.99	1.00
Bird	2.50	0.98
Rabbit	0.32	0.30

Age	Frequency	Severity
0-2	1.00	1.20
2-5	0.45	1.10
Above 5	1.30	1.00

- a)** Determine the number of risk cohorts is the data.

[1]

- b)** Which animal has the highest frequency and severity?

[1]

- c)** List the base cohort.

[1]

- d) Calculate the frequency & severity and burn cost for each risk cohort. [12]
- e) How does risk of a newborn Bird compare with that of a 3-year-old Rabbit? [1]
- f) Suggest other adjustments that should be made to the burn cost to arrive at the final price? [3]
- [19 Marks]**

Q. 10) You are a Pricing Actuary working for at Lloyds. The current Property exposure curve in use obsolete. You have been provided the below data:

Claim No	Claim Amount	Sum Insured
1234	60	600
2345	70	175
3456	150	750
4567	30	120
5678	100	200
6789	80	100
7900	120	160
9011	100	500
10122	80	100
11233	50	250

- a) Using above data, prepare an updated loss curve. [9]
- b) Calculate the burn cost for the following structures of the policy:
- No deductible or limits
 - Deductible of 75
 - Limit of 100
 - Deductible of 50 & limit of 100 [4]
- c) Discuss with reasons the best suitable option for insured and insurer amongst the above. [3]
- [16 Marks]**
